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In this video (6:20 minutes), Matt Woods and Professor Richard Eckard of the University of Melbourne discuss livestock emission reduction and avoidance.

MATT WOODS: Hello.

I'm Matt Woods.

And I'm here with Professor Richard Eckard. Richard has been working for over 20 years on addressing the impacts of a changing climate on agriculture.

A big part of the emissions story in Australia is livestock emissions. Can you briefly describe what they are and why they matter?

PROFESSOR RICHARD ECKARD: So the majority of livestock emissions come from what we call ruminants. Ruminants have got four stomachs. We've only got one. So ruminants are your cattle, sheep, goats; they would be your classic ruminants.

And because of the first stomach, so, a ruminant can actually digest cellulose; grass. You and I can't live on grass.

And they can because the first of their four stomachs called the rumen is a microbial digestion vat. So it's a massive vat with microbes that are there to break down the cellulose and turn it into sugars. So they can live on grass when you and I can't. But unfortunately, the moment you put organic material; grass, in an anaerobic environment, so no oxygen, you generate methane as well. We've known that for centuries because we know that when you put organic material or waste into an anaerobic pit, it'll generate a methane gas. Now unfortunately, that's what happens in ruminants as well. They generate this gas, methane. And methane is a powerful greenhouse gas. For every day that methane's in the atmosphere, it warms the planet a hundred times more than carbon dioxide. So it's important.

WOODS: Why is it important to farmers and farm businesses that they know this?

ECKARD: Well, 20 years ago, most farmers would not have even been aware that their cattle are producing methane.

But since climate change is now becoming quite a pressing issue, what we're seeing is not necessarily government policy, but we're seeing the supply chain starting to respond to the Paris Climate Agreement. And so all our multinational supply chain companies have started setting targets based on what we call the Science-Based Targets Initiative. So, you try to sell your meat to one of the major meat companies; by 2030, they'll be looking for 30 per cent less emissions from your meat. Because they're reporting down their supply chain to shareholders who are responding to customers.

And the major banks are on the same trajectory as well, where they are required to report to their - the Central European Bank, for example, on the emissions profile of their loan portfolios.

So, they're concerned about exposure to greenhouse gas emissions, and if you think of Australian agriculture, 70 per cent of what we produce is exported down these multinational supply chains.

And we know that by 2030, they are putting restrictions on what they will buy in terms of greenhouse gas emissions. So it's not got much to do with government targets. It's got to do with our supply chains that are actually wanting lower emissions supply in the future, which is why then it is of concern to farmers, whether they agree or not, it's in place. So we're in that phase right now of doing a lot of the research required to give farmers cost-effective solutions. That's actually really important.

But between now and 2030, raising awareness that you need to know your number; what is your emission, and what can you do about it between now and 2030? So that you remain at the head of the marketing queue when the supply chain come buying for low emissions product.

WOODS: Right. You mentioned cost effective things that farmers can do about it. What are some of those things?

ECKARD: So to answer that question, we need to understand how the supply chain can buy. You see, they can't buy on actual emissions, because then they just buy a small farm rather than a big farm, because a big farm produces more than a small farm. They can only buy on a unitary value, like emissions per unit of production, emissions per kilogram milk solids, emissions per kilogram of wool, or kilogram of meat - which is what we call emissions intensity.

So how do you shift the dial on emissions intensity? You just become more efficient. So you use current best practice of nitrogen use efficiency, fertiliser efficiency using legumes, grazing management, weaning rates.

The reason why a prime lamb operation has a lower emissions intensity. It's just because you get 150 per cent weaning, relative to a beef system that gets a bit somewhat below 100 per cent weaning. So you haven't changed the emissions, you've changed the denominator, which is how much live weight you're producing.

So getting that more efficient by reducing unproductive animal numbers, just current best practice will put you at the head of the queue by 2030.

WOODS: If I'm a farmer sitting here listening to this, and I'm a prime lamb producer, let's say. What's the first cab off the rank for me to try and bring my emissions down?

ECKARD: So your prime lamb is a good example in that you're focusing on a meat product, which means that you can get your weaning rates up to a 150 per cent. You can get them up to maybe, some exceptional producers, up to a 170 per cent.

That means that for every ewe that you have on the farm that is producing methane, you're now dividing it by 1.5 or 1.75 lambs that are coming out of the system that go down the supply chain. So that actually means that per unit of meat produced, you are lower emissions than your neighbour, who might be a 100 per cent weaning or 120 per cent.

So that can be achieved through better genetic selection, feed conversion efficiency, selecting for or reducing health costs... basically anything that reduces every day that the breeder herd is unproductive on the farm.

In this video (7:42 minutes), Matt Woods and Professor Richard Eckard of the University of Melbourne discuss feed quality and livestock emissions.

MATT WOODS: Hello. I'm Matt Woods, and I'm here with Professor Richard Eckard. Richard has been working for over 20 years on addressing the impacts of a changing climate on agriculture.

A part of methane emissions from ruminants is quality of feed that's going into the animal.

If I feed my prime lambs better quality feed, does that bring emissions down?

PROFESSOR RICHARD ECKARD: Most definitely. So, more so in Northern Australia than Southern Australia because in Northern Australia, the inherent range land is lower quality than some of our temperate grasses in the south. But no doubt if you're a lamb producer in Western Victoria and you bring more legumes into your pasture, the growth rate of the lambs will increase relative to just being a grass based pasture.

The same in Northern Australia, if you've got a beef system and you have a leucaena for example, or desmanthus or some of these these novel legumes, they would have two modes of action. One, they would improve the average quality of the grass, because a lot of the northern grasses are poor quality. So your growth rate would pick up from, say, 0.3 kilograms per day, to about 0.9 kilograms a day. So you're actually finishing animals for market earlier, but some of these legumes also have secondary compounds in them, like tannins that reduce methane for every day they're on the pasture. So you get two effects. One, better feed quality gets the animal growing faster so it gets out to market earlier, and that reduces your emissions intensity.

But the secondary compounds reduce methane for every day they're on the pasture, and that's a completely separate effect.

WOODS: Dairy farmers feed their cows pretty well. Are they going to produce less emissions per cow on a dairy farm than, say, a beef operation?

ECKARD: So here's the unfortunate irony, is that methane is very strongly related to dry matter intake.

And so you've got a Brahman in Northern Australia eating four, five kilos of dry matter a day because it's low quality; it takes longer time to go through the rumen. And you've got a dairy cow in Victoria that's producing 50 litres of milk at peak lactation on high quality pasture, eating 23 kilos of dry matter a day. So the dairy cow is producing more methane per kilogram, for total intake.

But per kilogram dry matter intake, they're actually no different. They're both producing about 20 times methane per kilogram dry matter intake.

The difference is that the emissions intensity of the dairy cow is much better.

Because it's eating 20 kilos, so it's producing more methane, but it's producing more product.

And so the emissions intensity of the dairy cow on better quality forage is much lower than say a beef cow in Northern Queensland growing at 0.3 kilos a day.

WOODS: Now are there some up and coming technologies that might be useful for farmers to to reduce some livestock emissions?

ECKARD: So there's a couple of things that farmers can do. We started off with the 'do now' stuff, which is, reducing unproductive animal numbers, animal health, breeding; breeding better, better genetics, better, faster gain.

There's a bunch of things that we can do around feeding animals.

Better legumes. There's a lot of legume technologies coming through, like desmanthus, leuceana. There's a lot of interest in using the current legumes in Southern Australia like white clover and lucerne, that we can actually express these secondary compounds more in those.

And so, in the pipeline coming through would be: yesterday, you bought white clover tomorrow you buy white clover with tannin in it that reduces methane as well as giving you a boost to production. So that's a space to watch, and it's an actively emerging space.

But stepping into the future, there are technologies like the seaweed option.

We've heard a lot about seaweed. There's another product, 3-NOP Bovaer, which does the same as seaweed.

Those are coming through in the marketplace. The problem is those methane inhibited products really only work where you've got confined animals, and you can feed them every day. Because you think of an inhibitor...

WOODS: So the dairy operation, for instance, it might be useful.

ECKARD: Yeah. In a dairy where you're feeding twice a day, that could work. If you're in a feedlot, it'll obviously work.

If you've got an inhibitor that only lasts for an hour in the rumen and before the rumen breaks it down. You think about that. It's gotta be in every mouthful of every mouthful the animal eats to be effective.

WOODS: Right. Yeah.

ECKARD: So, a feedlot works because you can mix it into every mouthful. And in a dairy might work because you can feed it twice a day, but it would be a lumpy effect.

WOODS: Yes, okay.

ECKARD: But in the end, we've got to move to something a bit more sustainable for the extensive grazing industries because these daily supplements - well, first of all, we don't want to doom farmers to have to pay a feed company every day for a supplement to reduce methane.

WOODS: Just out of interest, what are we talking about, sort of cost to a farmer?

ECKARD: The cheapest of these supplements for dairy would be about 50 cents per cow per day.

WOODS: Right. Okay.

ECKARD: And at the current carbon price of about 34 dollars a tonne, we're talking of seven cents per day as the pain point. So a farmer could afford seven cents a day if they're getting paid a carbon credit for that, and the price is 50; 50 cents.

WOODS: Right.

ECKARD: So we've got a way to go before methane supplements become a viable solution, which is why we're more interested in the legume technology because that's things we can do now. We have farmers using these legumes in across all the grazing industries.

So let's work with what we've got and at least we can get 20 per cent reduction in methane out of that.

But looking forward into the future, there are more speculative technologies coming through like early life programming.

Which is the concept that your and my gut microflora are a product of our upbringing, of our home environment. We inherit that.

It turns out the rumen could be the same. And there's a few papers published now showing that if you feed one of these inhibitors to cows and calves through pre-parturition, and through the birthing phase, and through the weaning phase that the animals can remain lower methane afterwards without supplementation.

And so that gives us hope that -

WOODS: Sorry, just to clarify that -

They get inoculated early in life, and then they wouldn't need any further supplementation.

ECKARD: Correct. That's the aim.

And so you start thinking of the Northern Territory where we don't even know where most of the cattle are, legume technology could work.

But establishing it on, you know, hundreds of square kilometres is a logistical issue.

But if you could come up with a once in a generation intervention, that resulted in all the calves being 20 per cent less methane, that's quite a big breakthrough.

It's not costing the farmer daily, you set them on the right course, and then they just are lower methane. So that has to be the ultimate goal of research, but it is still very speculative.

In this video (5:14 minutes), Matt Woods and Professor Richard Eckard of the University of Melbourne discuss soil carbon through grazing management.

MATT WOODS: Hello. I'm Matt Woods, and I'm here with Professor Richard Eckard.

Richard has been working for over 20 years on addressing the impacts of a changing climate on agriculture.

I'm going to move now to soil carbon and ask you about grazing management. Can you effectively build soil carbon through grazing management?

PROFESSOR RICHARD ECKARD: So there are a lot of claims around grazing management but the only actual evidence we have in the peer-reviewed literature would be that if you move out of a set stocking environment; a continuous grazing environment, where you might get caught out in a drought period will lead to erosion.

It will lead to the loss of soil organic matter off the surface of the soil through hoof action.

That we know; that we've seen in long term trials.

Once you move beyond that into some form of conservative rotational or adaptive grazing management strategy, there is no data to suggest that one would do soil carbon better than another.

They would all enhance soil carbon relative to the set stocked example.

So the best practice that we can advise farmers to do to build soil carbon in grazing systems is to move out of a set stocking environment into what we would call more adaptive rotational grazing.

Once you're into that environment of rotational grazing, there's no evidence that says one of the ten philosophies that are out there is better than another.

And that's as much as we can say right now.

WOODS: Ten philosophies, ten things to choose from. There must be a system that's going to work better for some farmers than others.

Is that right or is it just a bit of pot luck when they go to choose their system?

ECKARD: I think, it's fair to say that 40 years ago we thought there were recipes.

WOODS: Yep.

ECKARD: But then along came more variable climates and along came climate change. And that threw recipes out of the window because a recipe that you stick to religiously is going to get you in trouble when the climate varies.

So where we now say is all ten of those grazing philosophies have some merit at some time.

What we need to do is step back from fixed recipes to understand the principles of: if I overgraze or I have animals in too long and they regrazed the same plant twice in a row, they will graze the more productive plants first.

And so I'll get an increase in unproductive plants. And so your grazing management needs to be more adaptive than ever before.

So understanding the principles of grazing management, the need to rest those productive plants; those nutritious plants, the need to balance unproductive versus productive grasses, in the way you manage range land then becomes more critical.

And then you go to those ten philosophies and say, well, for this next three month period, because the El Niño's coming, I need to pick a different strategy. Because the one I'm doing will get me in trouble.

And so adaptive grazing says, understand the principles around how plants are productive and animals interact with the grassland system.

And then apply those principles in some form of conservative rotational resting, allowing plants to recover, allowing productive plants more time to recover, not allowing re-grazing of plants, all become important. But unfortunately, with a variable climate, we can't have fixed recipes.

WOODS: Sounds like what you're saying is that farmers need to be flexible in how they approach their grazing. Understand the concepts and be very flexible depending on what's happening.

Is that right?

ECKARD: Yeah. Most definitely, we need to be adaptive in our grazing management. The only thing we need to be fixed about, and some of the better farmers have figured this out, is have immovable and non-emotive cutoff dates.

So if you haven't got 25 millimetres by the 23rd of February, you need to lighten your stocking rate by x per cent.

Those are more critical rules that have emerged over time is: not having this emotion of "I'll just keep them for another week". Because that's how you get caught out.

WOODS: Okay.

ECKARD: The market's gone by then, and you've lost your opportunity.

So the better farmers have critical cutoff dates to make decisions.

But that decision is really still adaptive grazing. How do I adapt to this condition? I've got to unload a third of the animals and send them down to market, even though I might not get the best price, I now have the right stocking rate for the next three months.

In this video (8:39 minutes), New South Wales graziers Mike and Helen McCosker discuss adaptive grazing and soil health.

MIKE MCCOSKER: So the importance of soil carbon in understanding the health of my farm is critical. Soil carbon is the end result of all of the ecosystem functions working properly.

Whatever I'm producing relies on the green plants harvesting sunlight energy, turning that sunlight energy into sugars.

Soil carbon is the end result of all the ecosystems working properly.

So when we look at what's here now, the diversity above the ground indicates that I've got diversity of roots and diversity of biology under the ground. And that's the key to turning the sunlight energy into soil health and soil carbon.

So the greater the diversity of roots, the greater the diversity of biology, and that's what turns sunlight energy into soil carbon forming.

The penetrometer is a great tool to give me an indication of if there is a hardpan in this soil. So after years of planting oats every year and running the cattle across the country, then you know, there would be a hardpan somewhere in the first three or four inches. And the penetrometer shows me that as soon as I hit four hundred, the roots are no longer growing in that soil. So that shows me if the hardpan is just, you know, two inches or four inches under the soil.

So at the moment, I've got half a metre of good functional soil there.

And that means no hardpan for me, no restriction of root growth. I've got the all of the biology and the roots doing exactly what they're meant to be doing under there, and that includes taking in water and holding the water for the crop.

Soil carbon and the ability of this soil to take in water are directly proportional.

So when my soil carbon levels are low and the soil structure has collapsed, this soil can only take a maximum of, you know, 50 millimetres of rain in an hour. If I get more than 50 millimetres of rain in an hour, that water would hit the soil and run away.

When I get my soil carbon up, it makes the soil like a sponge. And so now that I could take in anything up to 500, 700 millimetres of rain in an hour. So I could get my whole annual rainfall in one hour, and it would all go into the soil. So the importance of soil carbon and the ability of the soil to take in water and make my rainfall now efficiently turn into crops and production directly proportional.

So grazing management for soil carbon and for diversity, it comes back to a planned rotation.

So I pull the cattle together so that my cattle impact when the cattle are in here. It's like a quick mow. So we're a short period of time with high stock density.

And then the other thing that I really plan is the rest period. So how long does this pasture need to then recover from that grazing?

So, we talk about time controlled grazing or planned grazing, but that's all about planning the rest period.

And to get the rest period, I plan the density of the stock and how long that land is impacted for.

I'm looking for indicator species. So I'll be looking for if this is one of my main grasses because it's nice and sweet and it's high in nutrition, then I'm looking for this grass to be fully recovered before I'm bringing those cattle back here.

This gets a little bit counterintuitive because when the growth is happening really quickly, I want to avoid, this grass having to draw from its root reserves to grow. So I'm wanting to avoid the animals coming back and taking a second bite before it's actually fully recovered and replenished the root system.

So when things are growing really quickly, I actually need to move the stock around the farm more quickly. When things are recovering slowly, I actually need a longer rest period because it takes longer for the plant to get away from where it's drawing on the root reserves back up to, what we call a late phase three or early phase four.

That so if things are recovering slowly because we're a bit short of moisture, we just haven't had the rain, then I'll actually slow the cattle down. There's no risk when things are running slowly that the animals will get a second bite, but there is risk that we don't get back to a fully recovered plant if I come back too quickly.

Does the soil carbon level show up as being different if we get dry weather? I would say absolutely.

The ability of the soil to absorb the water when it does rain means that my water cycle efficiency, that actually shows up when it stops raining because I'll have, you know, two times, three times more water held in my soil compared to the neighbour's place.

And, you know, so the pasture will stop growing on my neighbours and still be growing on my place simply because I've got more water held in the soil. 2018 and 2019 were horrendous years. They were bad for everyone.

I can say that we didn't need to feed the cattle for a lot longer into the drought before we had to start feeding.

And then when it did rain at the end of the drought, the land just recovered and regenerated so much more quickly. So the resilience of the farming operation, shows up in the extremes of the weather. When it's dry or even when it's particularly wet, having that soil carbon in the soil just gives this land much more resilience.

M. MCCOSKER: From the point of view of building soil carbon, you know we understand that it's not just one practice that makes that change. And in fact, I would say that we've stacked different practices together. We've changed the management of the cattle, and we've changed to a multispecies cover crops. And we've

HELEN MCCOSKER: And compost.

M. MCCOSKER: Included mineralised compost.

So we've stacked the changes.

When I think back about how the farm has changed, one of the things that is a little more subtle, but you do notice it really quickly, is that the soil gets softer.

And, it's soft when you walk on it. Yeah. It's easy for the tractor, so we actually probably now use less fuel.

And I noticed that because we also contract plant for other farms in the district.

And when we go on to another farm, we're pulling the same equipment, but we -

H. MCCOSKER: Gear

M. MCCOSKER: Gear slower.

H. MCCOSKER: Yeah.

M. MCCOSKER: And using more fuel.

H. MCCOSKER: Yep.

M. MCCOSKER: And then we come back home, and we go, wow. Isn't the soil so soft?

One piece of advice that I would give to people is hasten slowly, that don't be afraid to fail.

But if you do fail, make sure you've failed on just a little bit of the farm and work that out before you try and do it over the whole farm.

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